



main.py

Visualize

Run

Output



```

1 class RuleBasedSystem:
2     def __init__(self, facts, rules):
3         self.facts = set(facts)
4         self.rules = rules
5
6     def forward_chain(self):
7         iterations = 0
8
9         while True:
10             new_fact_added = False
11
12             for rule in self.rules:
13                 # Check if all conditions of the rule exist in current
14                 # facts
15                 if all(cond in self.facts for cond in rule['if']):
16                     if rule['then'] not in self.facts:
17                         print(
18                             f"Rule Triggered: IF {rule['if']} "
19                             f"THEN Add {rule['then']}"
20                         )
21                     self.facts.add(rule['then'])

```

```

Initial Facts: ['Socrates_is_human', 'All_humans_are_mortal']
Rule Triggered: IF ['Socrates_is_human', 'All_humans_are_mortal']
THEN Add Socrates_is_mortal
Final Knowledge Base Facts: {'Socrates_is_human',
'Socrates_is_mortal', 'All_humans_are_mortal'}

=== Code Execution Successful ===

```